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Installation survey

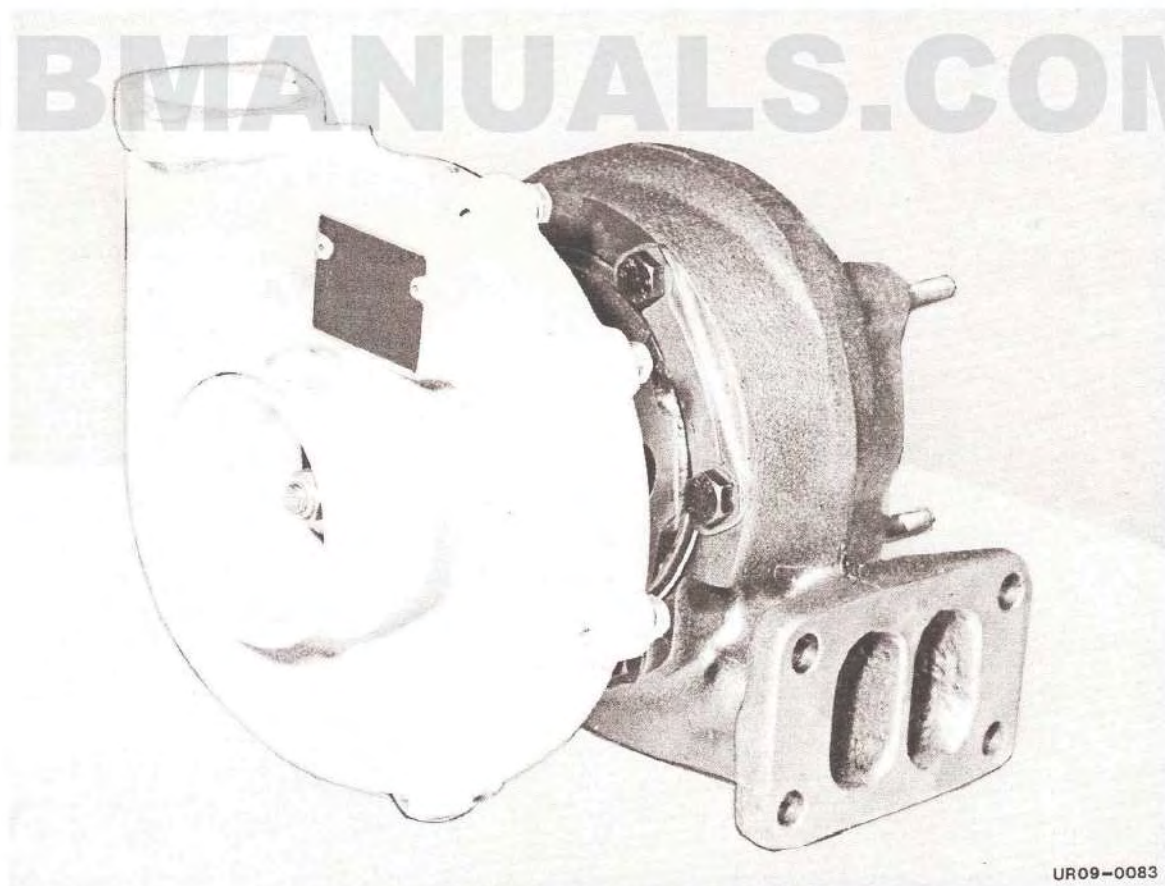
Chassis Model	Sales designation	Standard End chassis No.	Engine Model	Sales designation	Standard/SA End engine No.	Turbocharger Manufacturer	Type
435.110	U 1700	from 002 230	353.959	352 A. V	Standard from 577 967	Kühnle, Kopp & Kausch	3 LKs
.111	U 1700 ¹⁾	from 000 008			Standard from 413 968		
.115	U 1300 L ¹⁾	from PB ²⁾	353.959	352 A. V	SA 36 019 ³⁾ from PB ²⁾		
.117	U 1300 L/37						
.170	Thyssen-Henschel	from PB ²⁾	353.977	352 A. V	Standard from 575 178		

1) L = Truck

2) PB = Beginning of production

3) only valid for fire fighting vehicle

General view



UR09-0083

Settings

Max. end play of rotor shaft	mm	0.15
Max. radial play of rotor shaft	mm	0.65
Permissible runout of rotor shaft	mm	0.007

Special tools

Cons.No.	Tool	Number	Tool set
1	Torque wrench for shaft nut on compressor wheel	000 589 67 21 00	C
2	Dial gage to measure end and radial play	001 589 53 21 00	D

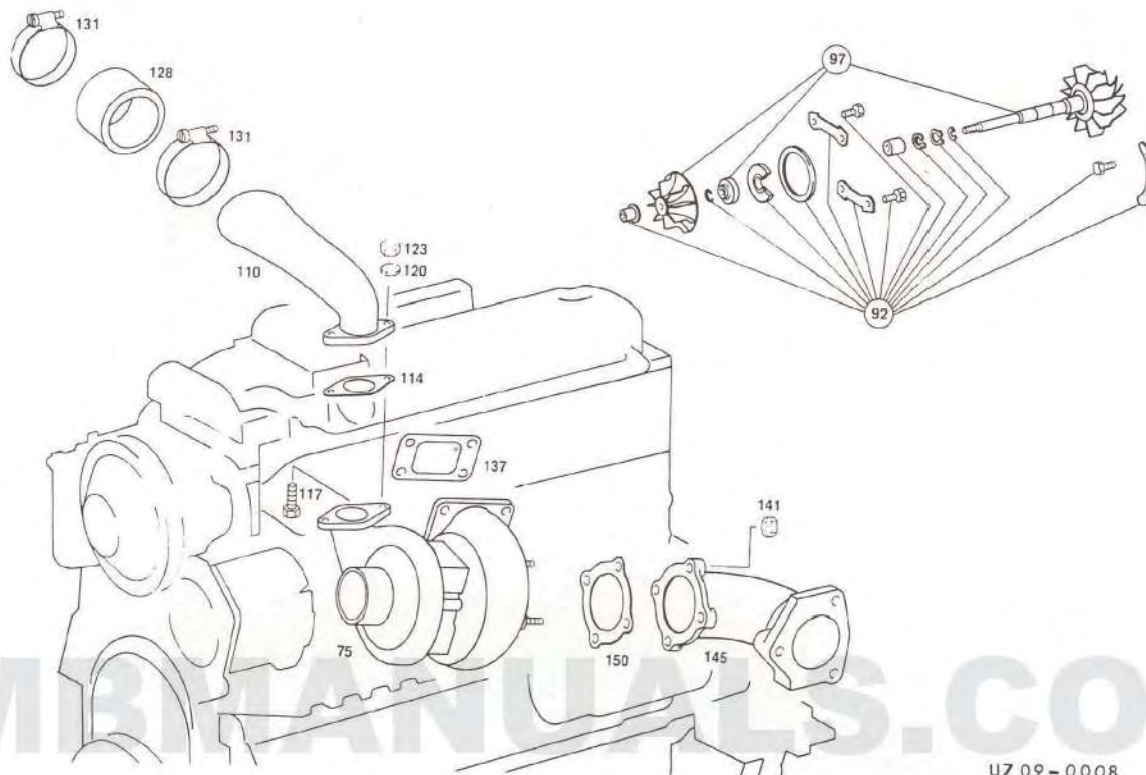
Necessary materials

Cons.No.	Description	Part No.
1	Curil T	001 989 37 20

Tightening torques

Location	Thread material	Nm
Shaft nut on compressor wheel		18
Compressor housing on bearing housing		6
Turbine housing on bearing housing		20
Flanged manifold to turbocharger	M 8	35 ± 5
Flanged manifold to exhaust pipe	M 10-8.8	45

Exploded view



UZ 09-0008

Turbocharger

75	Compressor	123	Nut
92	Repair set	128	Hose
97	Repair set	131	Expanding clip
110	Manifold	137	Gasket
114	Gasket	141	Nut
117	Bolt	145	Manifold
120	Lock washer	150	Gasket

Troubleshooting

Turbocharger

Fault	Cause	Remedy
Unusual exhaust fumes and reduced output	Air shortage due to inadequate charging pressure.	Check flange and hose connections of charging pipe for leakage.
	Interior of compressor housing dirty.	Remove and clean compressor housing. Check whether dirt is entering via center plate in cylinder head cover. If necessary, exchange cylinder head cover gasket. Tightening torque 25 Nm. Upon completion of cleaning, coat flange surface of compressor housing to bearing housing with 'Curil T', DB No. 001 989 37 20.
	Dirty air cleaner (negative pressure excessive upstream of compressor, with oil leakages at compressor end).	Clean air filter or exchange filter element. Check intake line. Check line downstream of turbocharger for leakage.
	Dirty or damaged exhaust line downstream of turbocharger or muffler.	Clean or recondition.
Unusually noisy	Leakage at joints and flanges of air and exhaust lines.	Check flanges and joints, exchanging gaskets if necessary.
	Rotor brushing against metal due to excessive bearing play.	Detach lines and check housing for brushing marks at turbine and compressor ends. If marks are found, exchange or recondition turbocharger.
	Bearings not being lubricated.	Guarantee correct oil pressure, cleaning or exchanging oil lines. If fault persists, overhaul turbocharger.
	Leakage at cylinder head cover.	Check gasket at center plate in cylinder head cover. Tighten cylinder head cover to correct torque setting. Exchange cylinder head cover if necessary. Tightening torque 25 Nm.

3 LKs

Notes on troubleshooting and installing turbocharger

Examining turbocharger in vehicle

Remove intake line at air inlet. Rotate impeller at turbocharger wheel nut, check that it rotates freely and evenly (rotate several times back and forward if necessary until the impeller is free of oil carbon deposits).

The impeller is centrifugally stabilized; it is not mounted in a fixed bearing. To check the impeller, move axially. If no rubbing on the turbine or turbocharger side is detectable, this means that the end play is okay. Deflect impeller radially and rotate. If no rubbing is detectable, the radial play is also okay.

Oil leaks at turbocharger

The oil-conveying bearing housing with piston rings is sealed from the air or exhaust-conveying spiral housing on all the designs of turbochargers we use. The function of this seal is based on the overpressure prevailing in the compressor housing. At elevated idling speed and when cruising, however, a vacuum may be produced favouring the escape of oil and moistening the charging paths with oil. The quantity of oil escaping is extremely small and has no measurable effect on the oil consumption of the engine.

If it is found that the charging air paths are moistened with oil when assembling, there is no reason to remove the turbocharger unless faults are determined when examining the turbocharger.

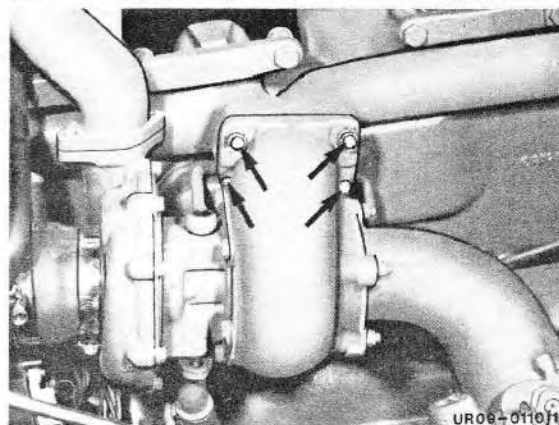
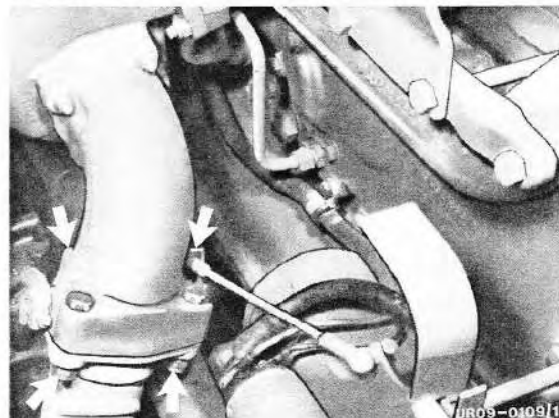
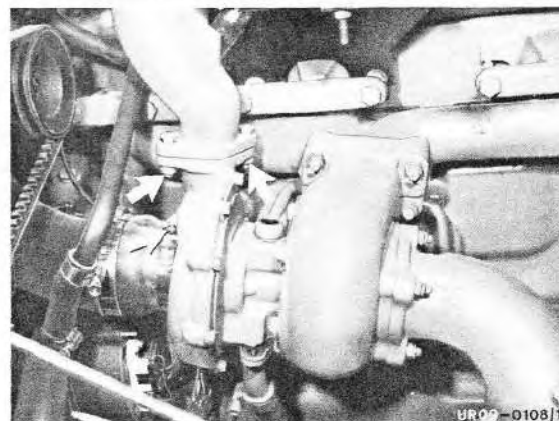
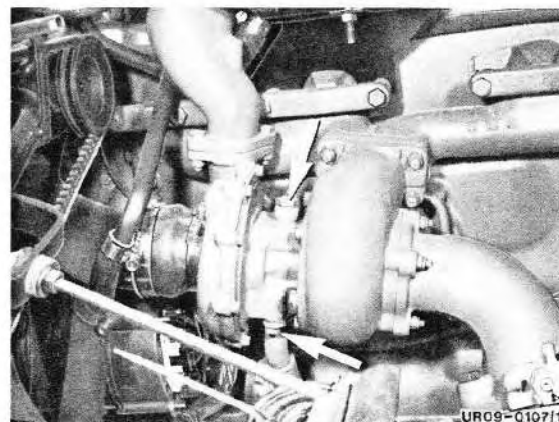
A further cause of oil leaks may be a damaged (deformed) or dirty oil return line. The consequent reduction in the line cross-section may increase the flow resistance in the oil return line causing the oil level in the turbocharger to rise and oil may escape via the seal at the turbine and compression end. In this case clean or exchange oil return line.

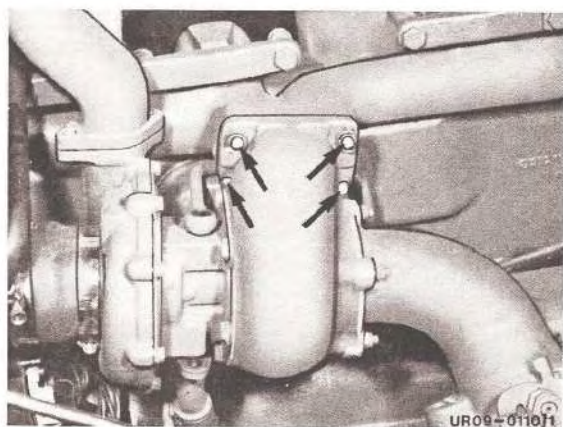
Assembling a replacement turbocharger

Before fitting the turbocharger, the intake line and flange manifold upstream of the compressor, the exhaust manifold and exhaust line as well as the oil feed and return lines must be very carefully examined for foreign bodies, dirt and damage. The seal caps at the openings of the turbocharger must only be removed shortly before installing. Before connecting the oil feed line, fill the bearing housing with clean engine oil through the oil inlet bore. Turn the rotating parts by hand to ensure that the bearings are covered by a film of oil.

Removal

- 1 Unscrew oil supply and return lines.
- 2 Detach air intake pipe and unscrew fastening bolts at flange.
- 3 Detach exhaust brake linkage and unscrew exhaust from turbocharger.
- 4 Unscrew and detach turbocharger at exhaust manifold.
- 5 Remove exhaust manifold.

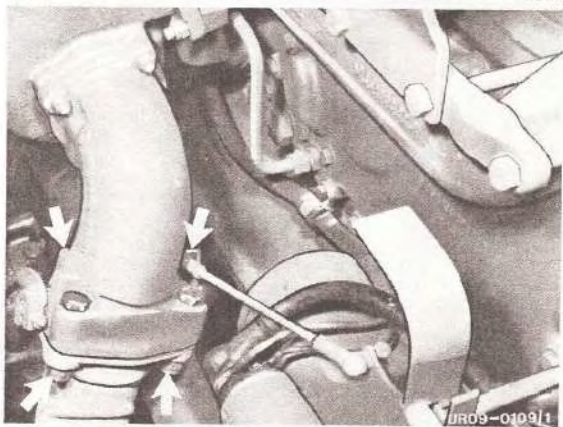




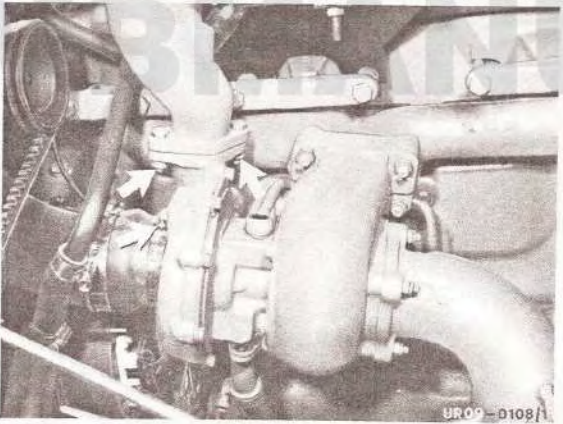
Installation

Prior to installing, check very carefully for dirt and damage the intake line upstream of compressor, flange manifold, exhaust line and exhaust manifold, as well as all oil lines.

1 Attach turbocharger to exhaust line with gasket.

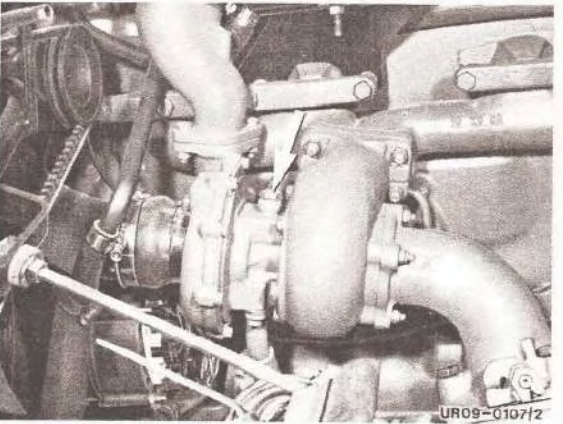


2 Attach turbocharger to exhaust with gasket. See 01.8—1.2/1 for tightening torque.



3 Tighten fastening bolts with gasket at flange, and attach air intake pipe.

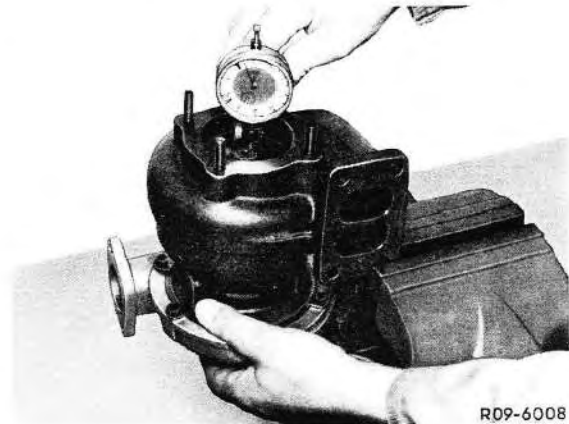
4 Bolt oil return line on.



5 Fill bearing housing of turbocharger with clean engine oil and connect oil delivery line.

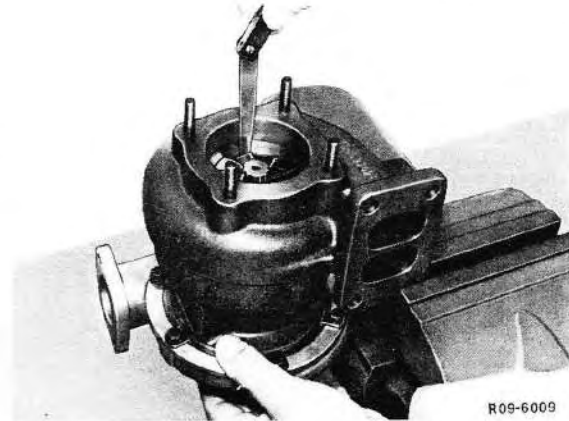
Checking

- 1 Remove turbocharger, referring to 2.1/1.
 - a) **Measure end play**
 - 2 Position measuring tip of special tool No. 2 on end of turbine rotor shaft.
 - 3 Press input shaft down, setting dial gage to "0".
 - 4 Press input shaft against special tool No. 2 and measure end play.
- See 1.2/1 for permissible end play.



- b) **Measure radial play** (to be determined at turbine end only).

- 1 Press turbine wheel to one side, using feeler gage to measure gap between turbine wheel and housing.
- 2 Push turbine wheel in opposite direction and measure gap with feeler gage.
- 3 Difference between two readings gives radial play.



Note: Measure at least at two different places.

Disassembly

- 1 Remove turbocharger, referring to 2.1/1.
- 2 Mark positions of turbine and compressor housings relative to bearing housing.
- 3 Remove fastening bolts, disconnecting compressor and turbine housings.



- 4 Clamp rotor at hub and unscrew shaft nut.

Note: Be sure not to damage rotor when removing shaft nut.

- 5 Remove compressor wheel.

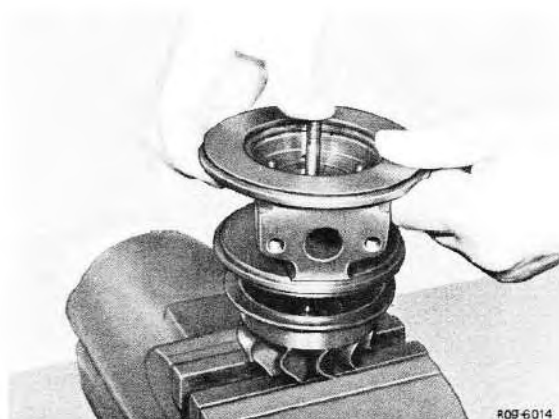


- 6 Remove snap ring.



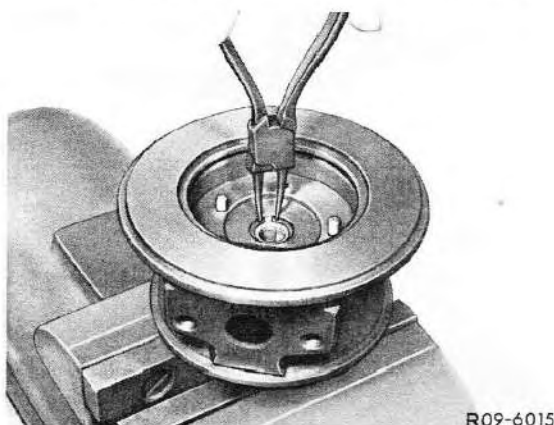
- 7 Remove bearing housing cover.
- 8 Push sealing bush out of bearing housing cover and remove piston rings.
- 9 Take out oil deflector, 1st bearing collar, thrust bearing bush and 2nd bearing collar.



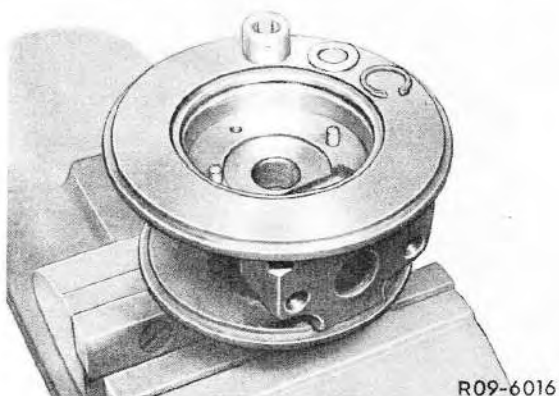


10 Take bearing housing off rotor.

11 Remove supporting ring and piston rings on rotor.



12 Take snap ring out of bearing housing at compressor end.

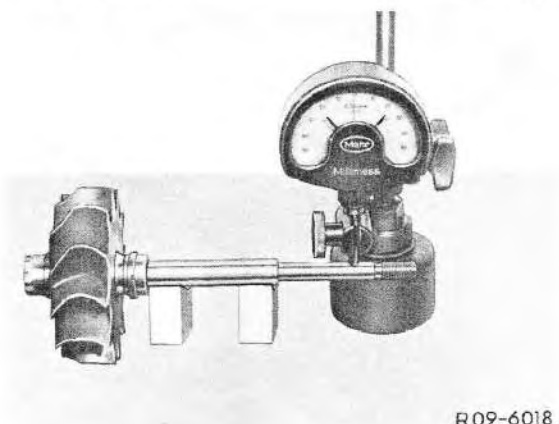


13 Take out bearing race and bearing member.

14 Remove snap ring.

15 Turn bearing housing over and remove both snap rings.

16 Clean and check all parts, exchanging if necessary.



Assembly

Note: Prior to installing each part, check carefully for cleanness.

1 Position rotor shaft on two V-blocks and check truth about 10 mm away from thread end, using special tool No. 1.
See 1.2/1 for permissible runout.

2 Install bearing member, bearing races and snap rings in bearing housing.

3 Clamp rotor at hub, install piston rings and place supporting ring in position.

- 4 Slip bearing housing over shaft.
- 5 Insert bearing collar, bush, thrust bearing, 2nd bearing collar and oil deflector.



- 6 Install sealing bush and piston rings in bearing cover.
- 7 Insert sealing ring into bearing housing, securing bearing housing cover with snap ring.

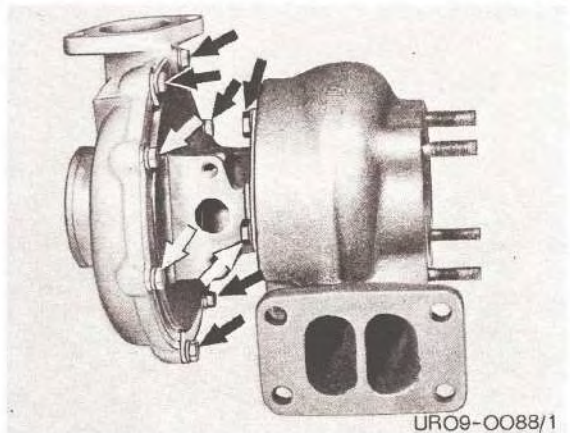


- 8 Place compressor wheel in position, torquing shaft nut to 18 Nm with special tool No. 1.
- 9 Check turbocharger shaft for ease of movement.
- 10 Note marks, coat compressor housing with sealant No. 1, attach to bearing housing and torque to 6 Nm.

Note: Coat bearing housing with sealant No. 1.



- 11 Attach segments and turbine housing with retaining plates to bearing housing, torquing to 20 Nm.
- 12 Turn turbocharger shaft and check for ease of movement.
- 13 Secure all bolts by bending metal tabs over.



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Installation Survey

Chassis Model	Sales designation	Series Chassis end No.	Engine Models	Sales designation	Series/SA Engine end No.	Turbocharger	
						Manufacturer	Type
435.110	U 1700	from 002 230	353.959	352 A. V/2	Series from 577 967	Al Research Garrett	To 4 B 27
.111	U 1700 L ¹⁾	from 000 008		352 A. LII/1 ⁴⁾	Series from 413 968		
.113	U 1700 L/38	from 084 642			Series from 693 577		
.115	U 1300 L ¹⁾	from PB ²⁾	353.959	352 A. V/2	SA 36 019 ³⁾ from PB ²⁾		
.117	U 1300 L/37	from 071 796		352 A. LII/1 ⁴⁾	SA 36 019 ³⁾ from 627 976		
.160	Attachment TM 170	from PB ²⁾	353.959	352 A. LII/1	Series		
.170	Attachment Condor	from PB ²⁾	353.977	352 A. V/2	Series from 575 178		

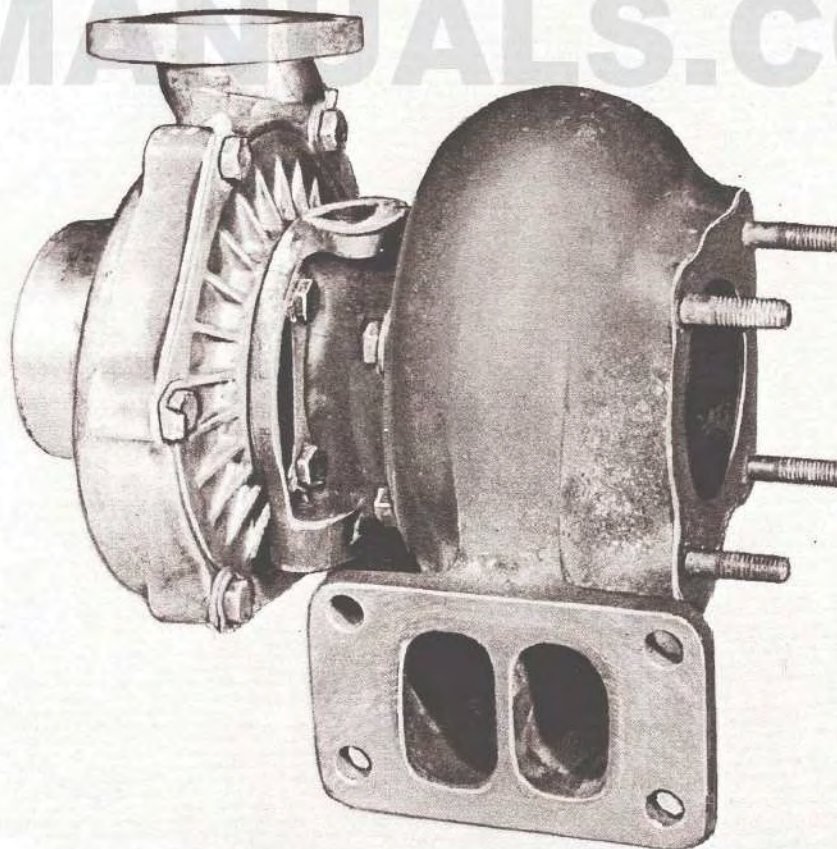
1) L = Truck

2) PB = Start of production

3) Valid only for fire services vehicles

4) From engine end No. 691 150.

General view



UR09-0089

Settings

Max. end play of rotor shaft	mm	0,10
Max. radial play of rotor shaft	mm	0,75
Permissible runout of rotor shaft	mm	0,007

Special tools

Cons. No.	Tool	Number	Tool set
1	Dial gage to measure end and radial play	001 589 53 21 00	D

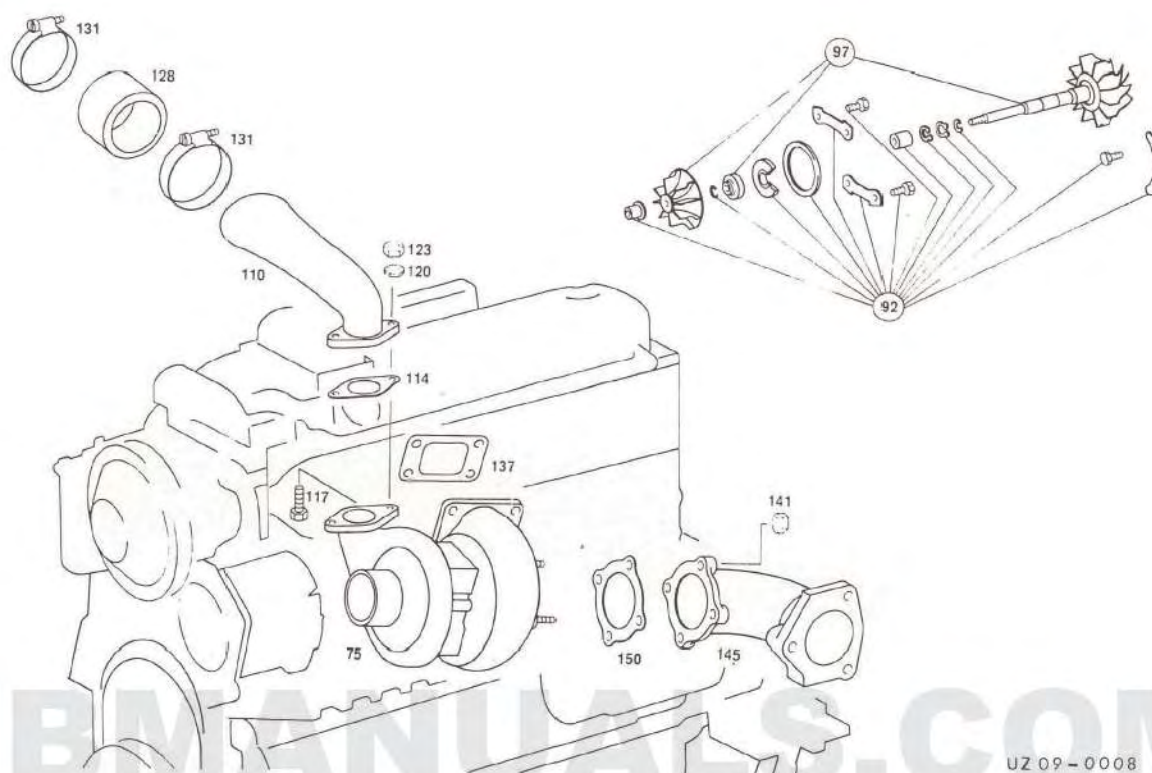
Necessary materials

Cons. No.	Description	Part No.
1	Curil T	001 989 37 20

Tightening torques

Location	Thread material	Nm	Torquing angle
Compressor back to bearing housing		8 to 10	
Fastening nut on compressor wheel	initial torque	21 to 23	
	final torque	—	90°
Bearing housing to turbine housing		11.5 to 15	
Compressor housing to bearing housing		11.5 to 15	
Flanged manifold to turbocharger	M 8	35 ± 5	
Flanged manifold to exhaust pipe	M10—8.8	45	

Exploded view



Turbocharger

75	Compressor	123	Nut
92	Repair set	128	Hose
97	Repair set	131	Expanding clip
110	Manifold	137	Gasket
114	Gasket	141	Nut
117	Bolt	145	Manifold
120	Lock washer	150	Gasket

Troubleshooting

Turbocharger

Fault	Cause	Remedy
Unusual exhaust fumes and reduced output	Air shortage due to inadequate charging pressure.	Check flange and hose connections of charging pipe for leakage
	Interior of compressor housing dirty.	Remove and clean compressor housing. Check whether dirt is entering via center plate in cylinder head cover. If necessary, exchange cylinder head cover gasket. Tightening torque 25 Nm. Upon completion of cleaning, coat flange surface of compressor housing to bearing housing with "Curil T", DB No. 001 989 37 20.
	Dirty air cleaner (negative pressure excessive upstream of compressor, with oil leakages at compressor end).	Clean air filter or exchange filter element. Check intake line. Check line downstream of turbocharger for leakage.
	Dirty or damaged exhaust line downstream of turbocharger or muffler.	Clean or recondition.
Unusually noisy	Leakage at joints and flanges of air and exhaust lines.	Check flanges and joints, exchanging gaskets if necessary.
	Rotor brushing against metal due to excessive bearing play.	Detach lines and check housing for brushing marks at turbine and compressor ends. If marks are found, exchange or recondition turbocharger.
	Bearings not being lubricated.	Guarantee correct oil pressure, cleaning or exchanging oil lines. If fault persists, overhaul turbocharger.
	Leakage at cylinder head cover.	Check gasket at center plate in cylinder head cover. Tighten cylinder head cover to correct torque setting. Exchange cylinder head cover if necessary. Tightening torque 25 Nm.

Notes on troubleshooting and installing turbocharger

Examining turbocharger in vehicle

Remove intake line at air inlet. Rotate impeller at turbocharger wheel nut, check that it rotates freely and evenly (rotate several times back and forward if necessary until the impeller is free of oil carbon deposits).

The impeller is centrifugally stabilized; it is not mounted in a fixed bearing. To check the impeller, move axially. If no rubbing on the turbine or turbocharger side is detectable, this means that the end play is okay. Deflect impeller radially and rotate. If no rubbing is detectable, the radial play is also okay.

Oil leaks at turbocharger

The oil-conveying bearing housing with piston rings is sealed from the air or exhaust-conveying spiral housing on all the designs of turbochargers we use. The function of this seal is based on the overpressure prevailing in the compressor housing. At elevated idling speed and when cruising, however, a vacuum may be produced favouring the escape of oil and moistening the charging paths with oil. The quantity of oil escaping is extremely small and has no measurable effect on the oil consumption of the engine.

If it is found that the charging air paths are moistened with oil when assembling, there is no reason to remove the turbocharger unless faults are determined when examining the turbocharger.

A further cause of oil leaks may be a damaged (deformed) or dirty oil return line. The consequent reduction in the line cross-section may increase the flow resistance in the oil return line causing the oil level in the turbocharger to rise and oil may escape via the seal at the turbine and compression end. In this case clean or exchange oil return line.

Assembling a replacement turbocharger

Before fitting the turbocharger, the intake line and flange manifold upstream of the compressor, the exhaust manifold and exhaust line as well as the oil feed and return lines must be very carefully examined for foreign bodies, dirt and damage. The seal caps at the openings of the turbocharger must only be removed shortly before installing. Before connecting the oil feed line, fill the bearing housing with clean engine oil through the oil inlet bore. Turn the rotating parts by hand to ensure that the bearings are covered by a film of oil.

Checking

- 1 Remove turbocharger.

Note: Remove and install with reference to 09.13–2.1/1.

a) Measure end play

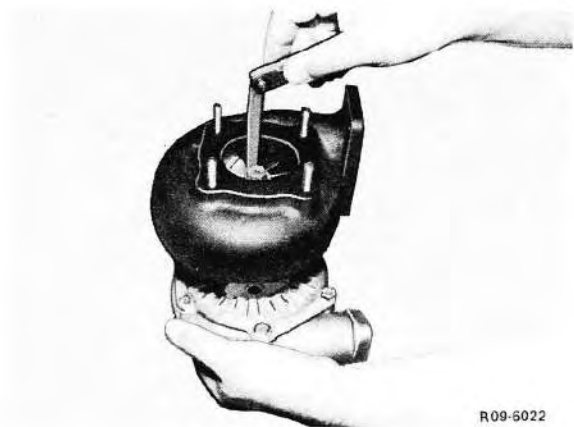
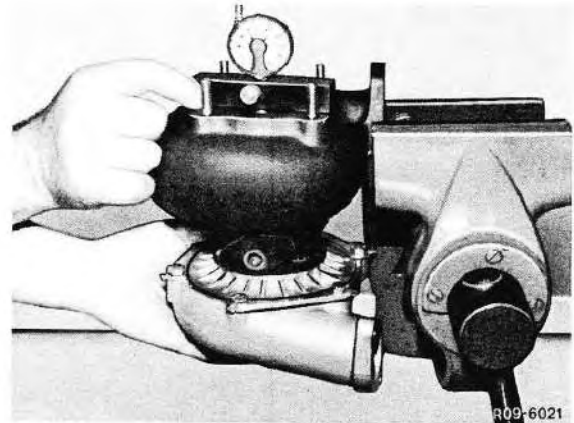
- 2 Position measuring tip of special tool No. 1 on end of turbine wheel shaft.
- 3 Press input shaft downwards, setting special tool No. 1 to "0".
- 4 Push input shaft against special tool No. 1 and measure end play.
See 1.2/1 for permissible end play.

b) Measure radial play

(to be determined at turbine end only)

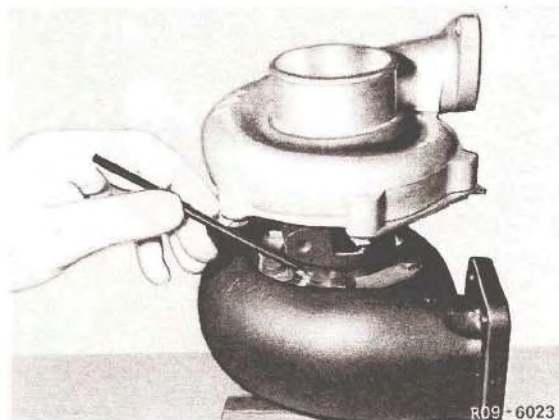
- 1 Press turbine wheel to one side, using feeler gage to measure gap between turbine wheel and housing.
- 2 Push turbine wheel in opposite direction and measure gap with feeler gage.
- 3 Difference between two readings gives radial play.
See 1.2/1 for permissible radial play.

Note: Measure at at least two different places.



Disassembly

- 1 Remove turbocharger, referring to 09.13–2.1/1.
- 2 Mark positions of turbine and compressor housings relative to bearing housing.
- 3 Remove fastening bolts, disconnecting compressor and turbine housings.



- 4 Introduce bearing housing in fixture at turbine end, removing shaft nut.

Note: You are requested to make the fixture in your own workshop with reference to drawing on page 4.1/4.

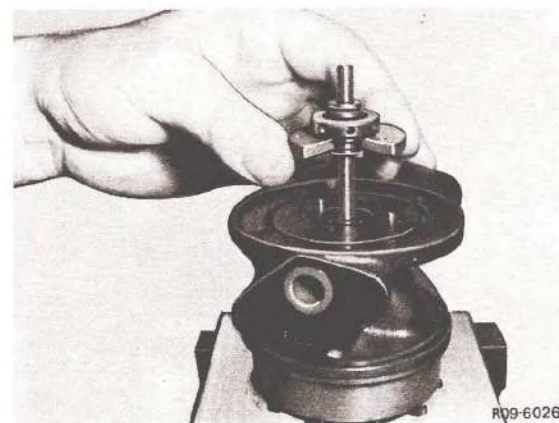
- 5 Remove compressor wheel and turn bearing housing.



- 6 Remove retainer and bolts from compressor back, taking compressor back off.



- 7 Remove thrust ring and thrust bearing.





R09-6027

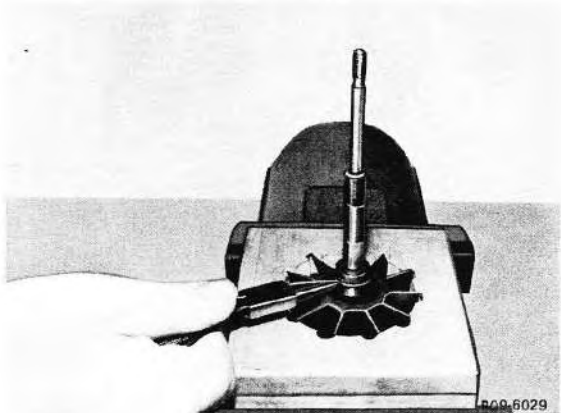
- 8 Remove sealing ring and take piston ring off thrust ring.



R09-6028

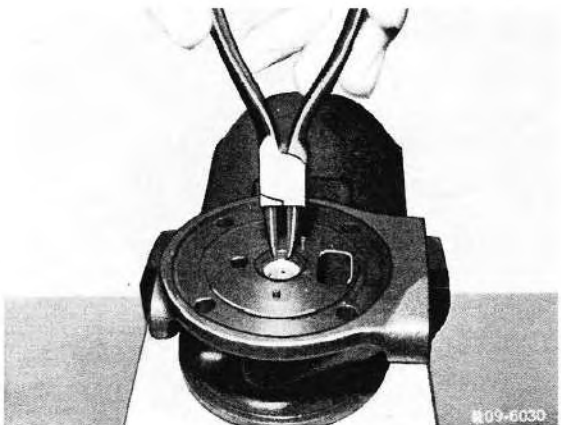
- 9 Draw bearing housing off shaft.

- 10 Remove shield.



R09-6029

- 11 Take piston ring off shaft at turbine end.



R09-6030

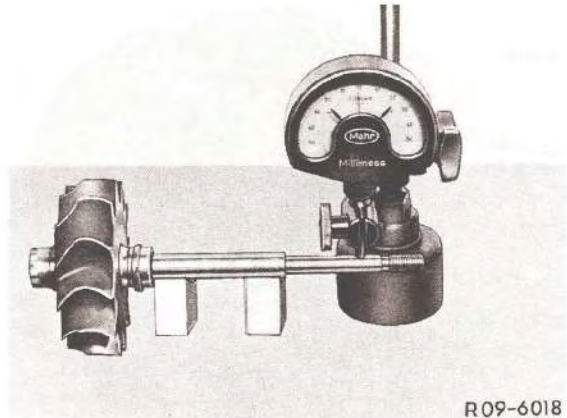
- 12 Remove both radial bearings and snap rings from bearing housing.

- 13 Clean and check all parts, exchanging if necessary.

Assembly

Note: Ensure each part is thoroughly cleaned before assembly.

1 Place rotor shaft on two V-blocks and check for concentricity approx. 10 mm away from the threaded end using a dial gauge.
See 1.2/1 for permissible runout.



R09-6018

2 Insert inner retaining rings in bearing housing, introduce radial bearing and fit outer retaining rings.

3 Insert turbine wheel and shaft in fixture.



R09-6032

4 Place piston ring on thrust ring, position thrust bearing on thrust ring and slip over charger shaft so that the thrust bearing rests on the bearing housing and the locking pins are correctly located in the bearing housing. Fit sealing ring in bearing housing.

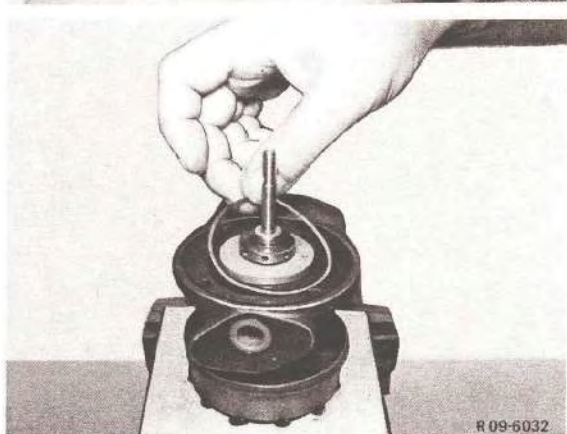


R09-6031

5 Insert gasket in bearing housing.

6 Slip back section of compressor together with the compression spring over the charger shaft and thrust ring, attach to bearing housing with retainers and tighten to 8 to 10 Nm (note marking!)

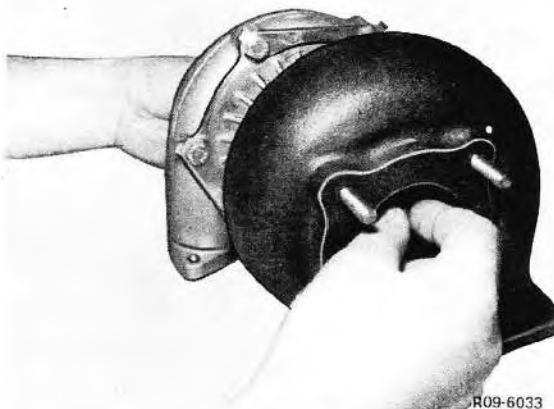
Note: When fitting back section of compressor, ensure piston ring is not damaged



R09-6032

09.14 Disassembly and Assembly of Turbo Charger

To 4 B 27



R09-6033

7 Place compressor wheel in position and tighten fastening nut in two stages.

Tightening torques:

1. 1st stage: initial torque 21 to 23 Nm
2. 2nd stage: final torque: 90°

8 Turn charger shaft, checking for ease of movement.

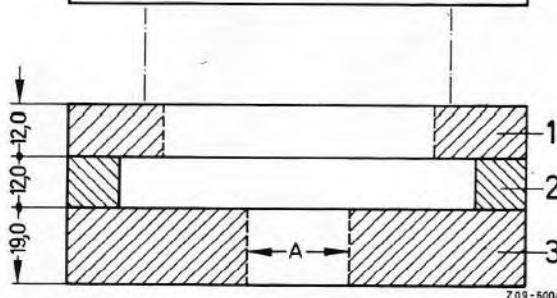
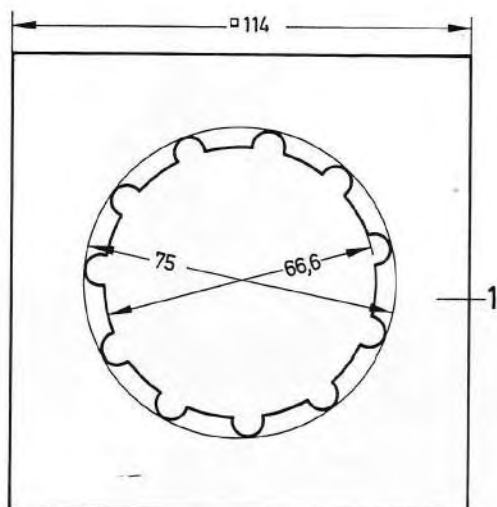
9 Mount bearing housing with retainers and shims in turbine housing and tighten to 11,5 to 15 Nm (note marking).

Note: Before assembling, coat bolt threads with sealing compound No. 1.

10 Attach compressor housing to bearing housing with retainers and shims, tightening to 11,5 to 15 Nm (note marking).

11 Check for each of movement of the turbocharger by turning the shaft.

12 Secure all bolts by bending metal tabs over.



Z09-9004

Turbine Wheel Fixture

Made of plywood, hardwood or aluminium

- 1 Upper section
- 2 Spacer
- 3 Base
- 4 Cutout for end of turbine wheel

Survey

Version

Air compressor

Additional air compressor

13.8

Additional air compressor

13.9

Belt drives

Belt drives

13.14

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Installation Survey

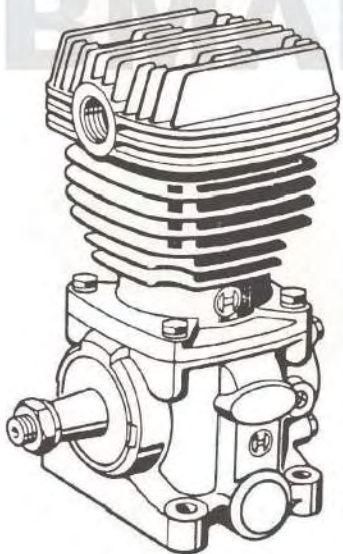
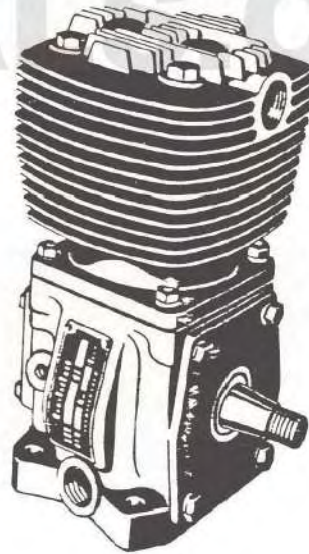
Chassis Model	Sales designation	Compressor Manufacturer Part No.	Version	Installation
435.110	U 1700	Bosch ²⁾ 003 131 13 01	Single cylinder compressor 125 cm³	Series from PB ³⁾
.111	U 1700 L ¹⁾			
.113	U 1700 L/38			
.115	U 1300 L ¹⁾			
.117	U 1300 L ¹⁾ /37	Westinghouse ²⁾ 003 131 5001	Single cylinder compressor 159 cm³	
.160	Attachment TM 170			
.170	Attachment Condor			

1) L = Truck

2) Optional

3) PB = Start of production

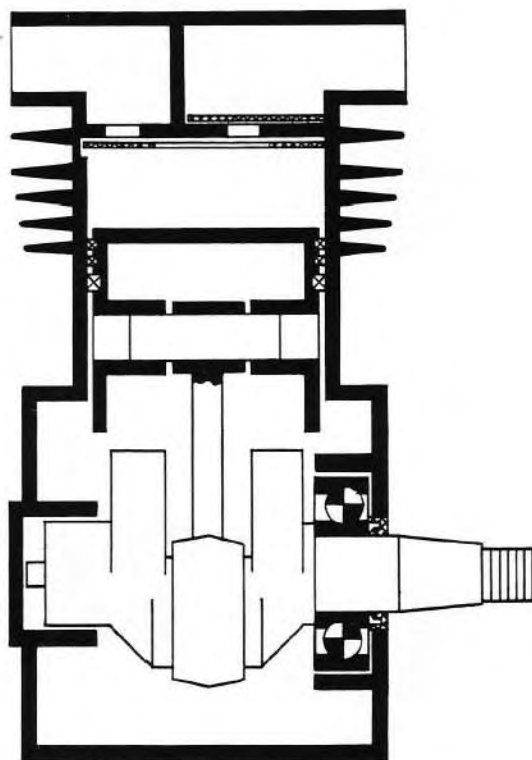
General view

Compressor 125 cm³ – BoschCompressor 159 cm³ – Westinghouse

Technical Data

Manufacturer		Bosch	Westinghouse
Type of lubrication		Circulation lubrication with oil delivery connection(UD)	
Piston-ø	mm	60	75
Stroke	mm	44	36
Displacement	cm ³	125	159
Maximum operating speed	1/min	3000	
Maximum speed (short duration)	1/min	3500	3300
Operating pressure	bar	18,5±0,3	
Maximum pressure (short duration)	bar	20,0	
Minimum required cooling air velocity	m/s	6,0	
Weight	ca. kg	8,2	10,0

Sectional view



UZ43-0208

Air compressor

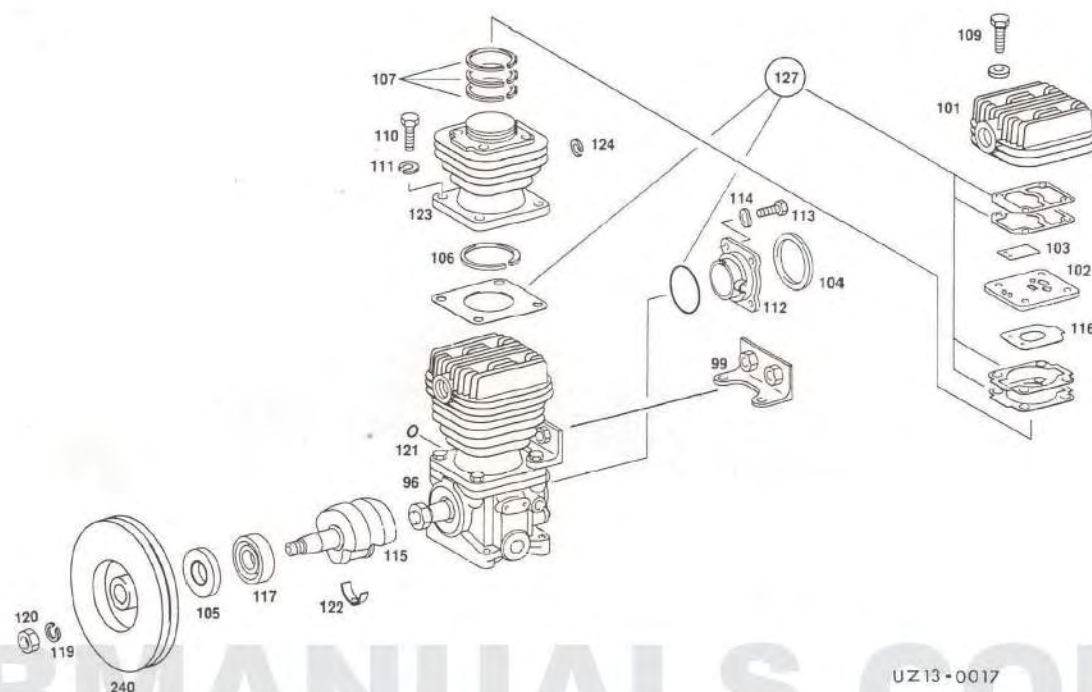
Tightening torques

Designation	Thread	Nm
Air compressor to bracket	M 8	22
Bracket to engine support		75
Cylinder head bolts ¹⁾	M 8	35
V-belt pulley to crankshaft	M 18	80 to 100

1) Insert cylinder head bolts dry, tighten to specified tightening torques and after warming up **slightly release the bolts individually** when cold and tighten to the same tightening torque.

During the next servicing interval, once again tighten the cylinder head bolts with the air compressor cold as previously described.

Exploded view



UZ13-0017

Additional air compressor

- | | |
|---------------------|--------------------------------|
| 96 Compressor | 113 Screw |
| 99 Bracket | 114 Spring washer |
| 102 Screening plate | 115 Crankshaft |
| 103 Spring | 116 Valve disc |
| 104 Seal ring | 117 Ball bearing |
| 105 Seal ring | 119 Lock washer |
| 106 Piston ring | 120 Nut |
| 107 Piston rings | 121 Seal ring |
| 109 Screw | 122 Bearing shell |
| 110 Screw | 123 Cylinder liner with piston |
| 111 Lock washer | 124 Lock ring |
| 112 Bearing cap | 127 Repair kit |
| | 240 Pulley |

Contents

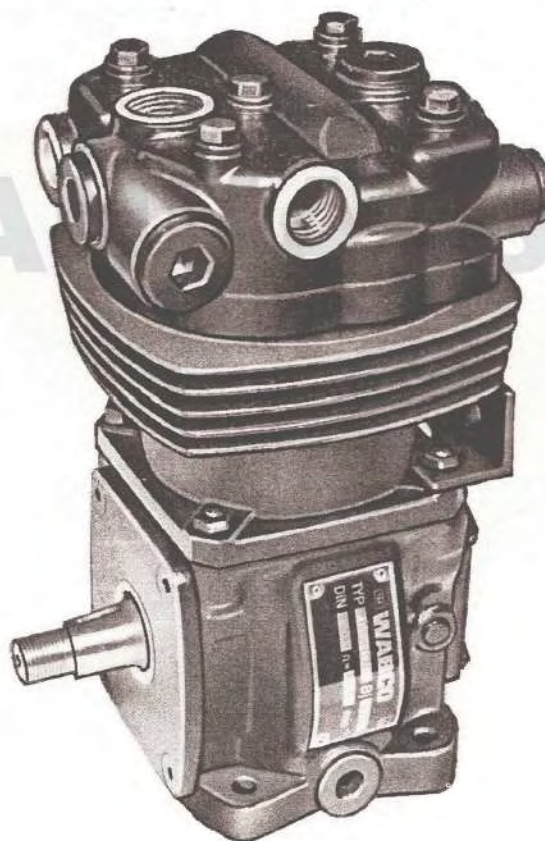
Chapter	Page
1 General	
Installation survey	1.1/1
General view	1.1/1
Technical data	1.2/1
Sectional view	1.2/1
Tightening torques	1.2/1
Exploded view	1.3/1

Installation survey

Chassis Model	Sales designation	Air compressor Manufacturer Part No.	Version	Installation
435.110	U 1700	Westinghouse 003 131 12 01	Single air compressor 229 cm ³ connected to the coolant circuit	SA 35 902
.111	U 1700 L ¹⁾			
.115	U 1300 L ¹⁾			
.117	U 1300 L ¹⁾ /37			

1) L = Truck

General view



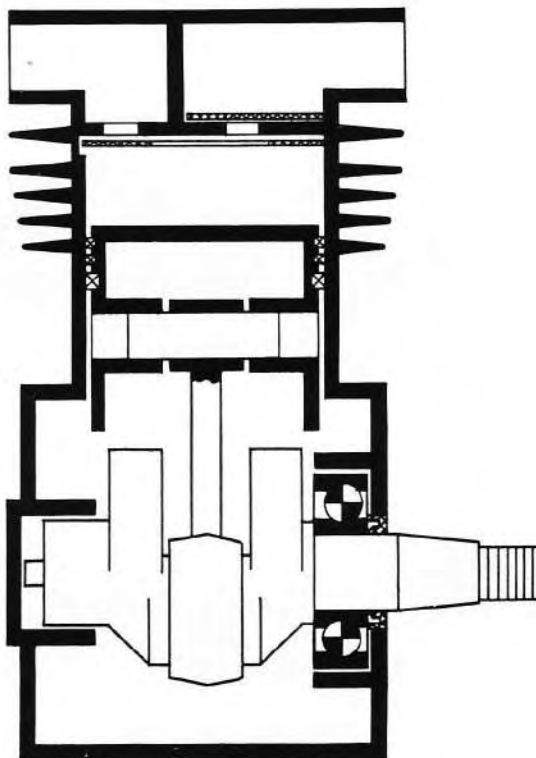
UR13-0061

Air compressor 229 cm³

Technical Data

Version		Single cylinder compressor
Cooling System		Connected to the coolant circuit of engine
Minimum required quantity of coolant	l/min	4,0
Type of lubrication		Circulation lubrication with oil delivery connection (UD)
Piston diameter	mm	90
Stroke	mm	36
Displacement	cm ³	229
Maximum operating speed	1/min	2800
Maximum speed (short duration)	1/min	3000
Operating pressure	bar	18,5±0,3
Maximum pressure (short duration)	bar	20,0

Sectional view



UZ43-0208

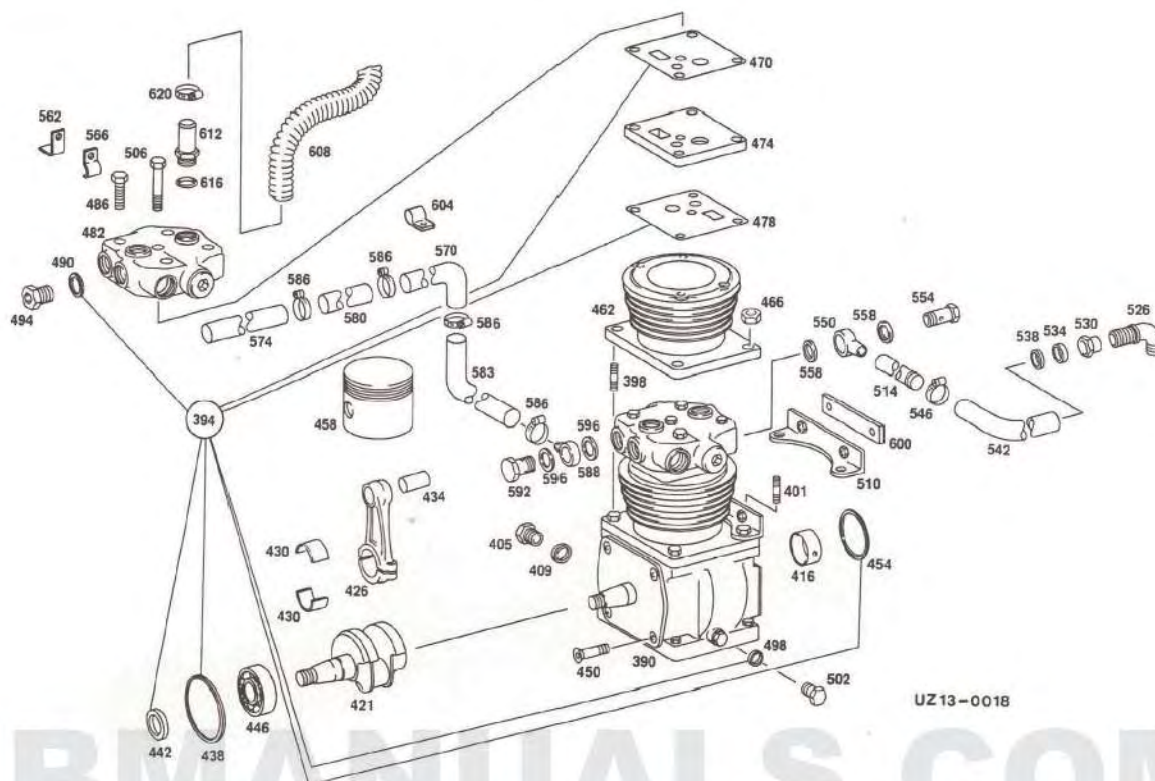
Air compressor
Tightening torques

Designation	Thread	Nm
Air compressor to bracket	M 8	22
Bracket to engine support		75
Cylinder head bolts ¹⁾	M 8	27 to 30
Valve plate to cylinder head	M 8	34
V-belt pulley to crankshaft	M 16	100±10

1) Insert cylinder head bolts dry, tighten to specified tightening torque and after warming up, **slightly release bolts individually** when cold and tighten to same tightening torque.

During next service interval, once again tighten the cylinder head bolts with the air compressor cold as previously described.

Exploded view



Supplementary air compressor

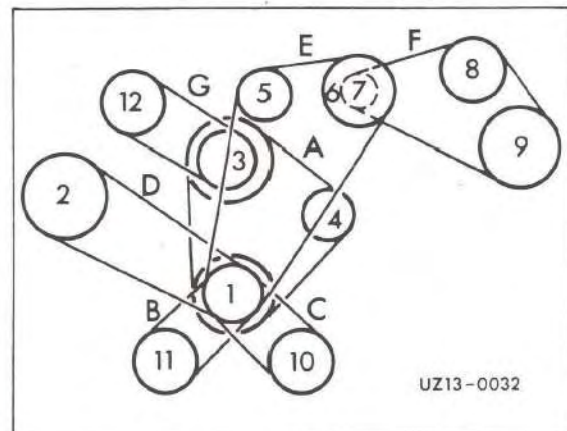
390	Compressor	470	Gasket	554	Hollow screw
394	Repair kit	474	Valve plate	558	Seal ring
398	Stud	478	Gasket	562	Bracket
401	Stud	482	Cylinder head cap	566	Clamp
405	Screw plug	486	Screw	570	Return line
409	Seal ring	490	Seal ring	574	Line
416	Bushing	494	Screw plug	580	Hose
421	Crankshaft	498	Seal ring	583	Hose
426	Connecting rod	502	Screw plug	586	Clamp
430	Bearing shell	506	Screw	588	Connector
434	Bushing	510	Bracket	592	Hollow screw
438	Seal ring	514	Line	596	Seal ring
442	Seal ring	526	Fitting	600	Bracket
446	Cylinder roller bearing	530	Nut	604	Clamp
450	Screw	534	Seal ring	608	Hose
454	Seal ring	538	Thrust ring	612	Pipe line
458	Piston	542	Hose	616	Seal ring
462	Cylinder liner	546	Clamp	620	Clamp
466	Nut	550	Connector		

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V-belt — Fan drive	2.2/1
V-belt — Air compressor	2.3/1
V-belt — Power steering pump	2.4/1
V-belt — Hydraulic pump	2.5/1
V-belt — Coolant pump/Generator	2.6/1

V-Belt Arrangement

No.	Pulley	Diameter mm
1	Crankshaft	200 ¹⁾ /187
2	Air compressor	185
3	Coolant pump	194 ²⁾
4	Alternator	98 64 ³⁾
5	Tension roller	97
6	Intermediate bearing	141
7	Intermediate bearing	96
8	Tensioning roller	97
9	Fan	126
10	Hydraulic pump (steering)	164 138,5 ⁴⁾
11	Hydraulic pump	153
12	Compressor-air conditioning	143



- 1) Only for v-belt "A", coolant pump-alternator.
 2) IVm. air conditioning system (SA 35 914) dia 196 drive and dia. 153 drive for compressor.
 3) Alternator N1-28 V 55 A 25 (SA 35 496).
 4) Model 435.115
 .117.

V-belt	Application	Dimensions in mm	Amount of yield in mm
A	Coolant pump – alternator	12,5 × 1250 ¹⁾	0 to 5
		12,5 × 1275	
B	Hydraulic pump	12,5 × 925	5 to 10
C	Power steering pump	12,5 × 950 12,5 × 955 La. ²⁾	
D	Air compressor	12,5 × 1350 ³⁾ 12,5 × 1425	
E	Intermediate drive – fan	12,5 × 1625 ²⁾ 12,5 × 1700	
F	Fan drive	12,5 × 1125	
G	Compressor-air conditioning system (SA 35 914)	12,5 × 1075 La	

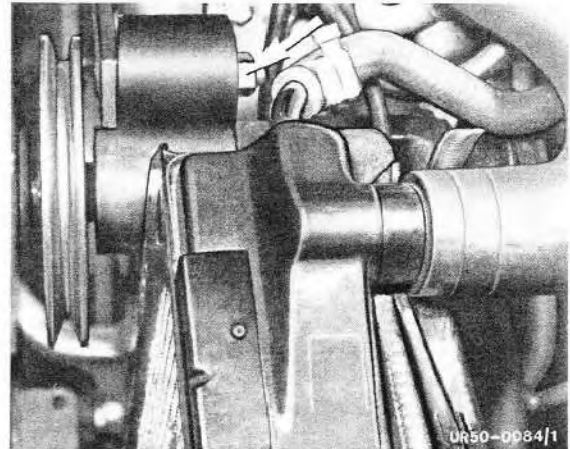
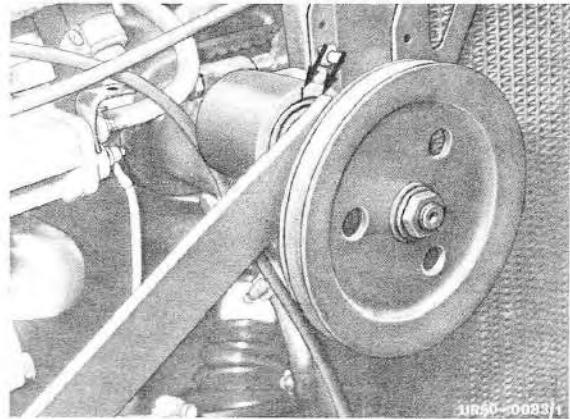
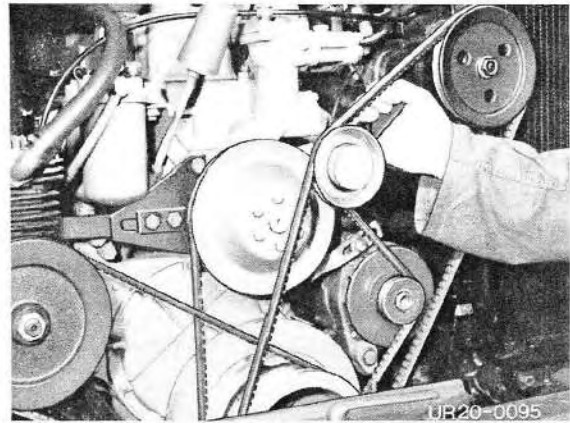
- 1) Model 435.115 special vehicle with alternator N1-28 V 55 A 25.
 2) Model 435.115
 .117.
 3) From chassis end No. 000 224

Tightening torques

Location	Thread	Nm
Air compressor to bracket	M 8 × 50	22

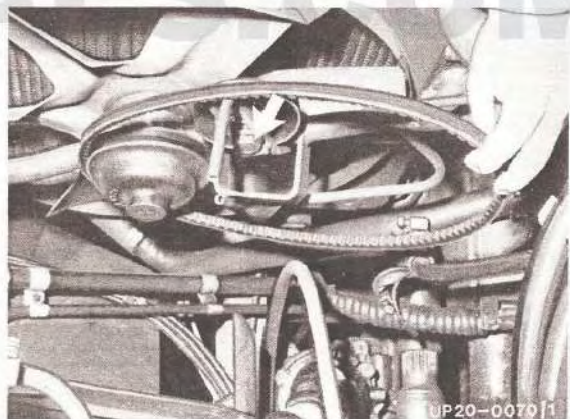
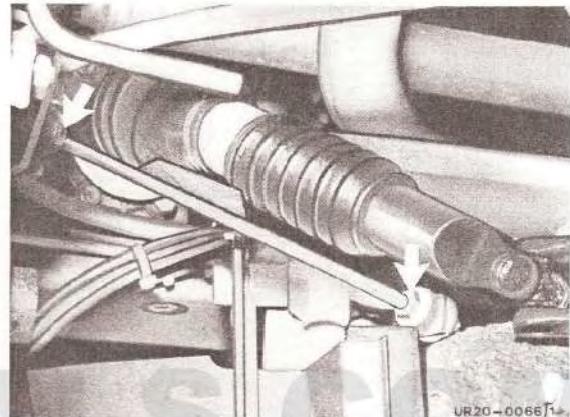
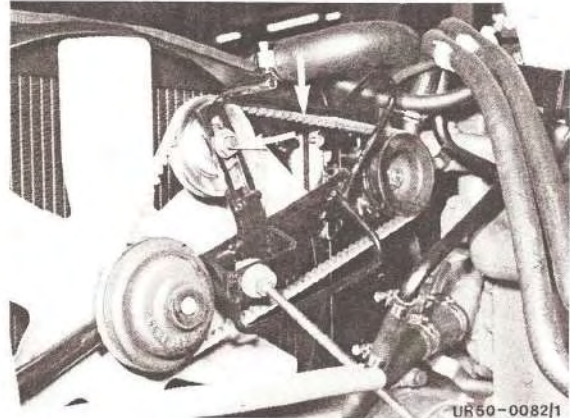
V-belt, intermediate fan drive

- 1 Prop up engine hood and remove front panel.
- 2 Check V-belt for condition and tension, exchanging if necessary.
See 1.1/1 for amount of yield.
- 3 To exchange V-belt, force tension bearing downwards and remove V-belt.
- 4 Check tension bearing adjustment, making sure that marks agree when V-belt is off.
- 5 Should marks not agree, slacken fastening nut at back and shift tension bearing in teeth by appropriate amount. Finally tighten fastening nut again.
- 6 Install in reverse sequence.



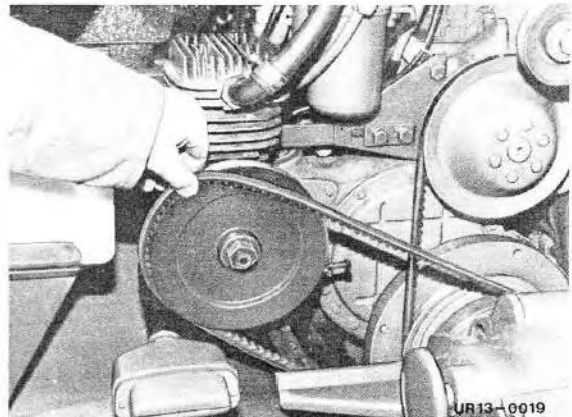
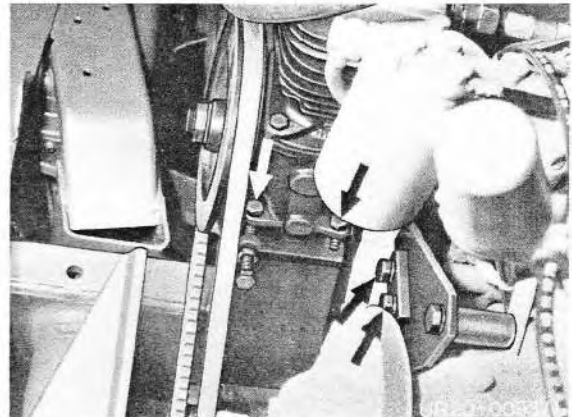
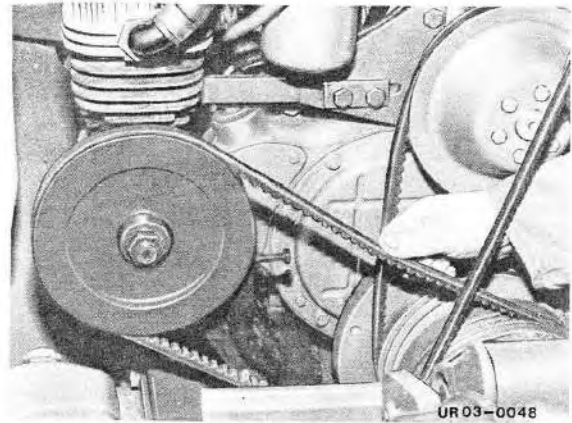
V-belt, fan drive

- 1 Remove engine hood.
- 2 Check V-belt for condition and tension, tightening or exchanging if necessary.
See 1.1/1 for amount of yield.
- 3 To exchange V-belt, remove radiator support.
- 4 Release jockey roller and detach V-belt.
- 5 Install in reverse sequence.



V-belt, air compressor

- 1 Prop up engine hood and remove front panel.
- 2 Check V-belt for condition and tension, exchanging if necessary.
See 1.1/1 for amount of yield.
- 3 Remove V-belt of intermediate fan drive, referring to 2.1/1.
- 4 Release air compressor mounting and tensioning screw.
- 5 Remove V-belt.
- 6 Install in reverse sequence, referring to 1.1/1 for tightening torque.



V-belt, power steering pump

- 1 Remove front panel and prop up engine hood.
- 2 Check V-belt for condition and tension, tightening or exchanging if necessary.
See 1.1/1 for amount of yield.
- 3 Remove V-belts of intermediate fan drive and air compressor, referring to 2.1/1 and 2.3/1.
- 4 Release pump mounting and slacken tensioning screw.
- 5 Remove V-belt.
- 6 Install in reverse sequence.

